8 Features of manufacturing industries

8.3 Modern manufacturing methods and systems when designing for manufacture for small, medium and large-scale production

Topic	What students need to learn:
8 Features of manufacturing industries	8.3 – Characteristics, processes, application, advantages and disadvantages and the importance of considering accuracy of production and efficiency of modern manufacturing methods and systems when designing for manufacture for small, medium and large scale production: *a) production scheduling and production logistics* *b) robotics in production – robots on fully-automated production and assembly lines/cells*

Topic	What students need to learn:
8 Features of manufacturing industries	*c) materials handling systems – automated storage and retrieval systems (ASRS), automatic guided vehicles (AGVs)* *d) flexible manufacturing systems (FMS), modular/cell production systems* *e) lean manufacturing using just-in-time (JIT) systems* *f) standardised parts, bought-in components* *g) quick response manufacturing (QRM)* *h) data integration – product data management (PDM), enterprise resource planning (ERP) systems* *i) concurrent manufacturing.*

f) Standardised parts, bought-in components

Standardised Parts and Components

- **Standardised parts** and components can be relied upon to fit correctly when adding or replacing elements of a design, such as a refill in your ball point pen or punched A4 paper in your ring binder.
- Screw fasteners are one of the most important examples of standardisation and most of the world now uses a metric ISO screw thread system.
- Light bulbs and fittings are another example of standardised components. Bayonet fittings, e.g. are regulated by BS EN 61184, which dictates their dimensions, tolerances and other requirements.

Bought-in Parts and Components

Standardisation ensures **bought-in parts** and **components** will ***integrate*** seamlessly in a manufacturer's product's. Examples of bought-in parts and components include:

- *Tyres*
- *Audio connectors*
- *Batteries*
- *Fuses*
- *Printer cartridges*
- *Ball-point pen refills*
- *Memory cards*

Bought-in Parts and Components

The benefits of using bought-in components include:

- *The availability of sub-assemblies (e.g. electric motors, hydraulic valves and gearboxes)*
- *Suppliers make standardised CAD files available for easy integration into CAD designs*
- *The components can be bought in bulk, which considerably reduces the unit cost*
- *There is no need to have their own expensive and potentially time-consuming manufacturing facility*
- *The specialist equipment and expertise of suppliers is not required in their own factory*
- *Specialist manufacturers work to specified tolerances for high quality*
- *The ISO 9000 accreditation framework can be used to maintain quality standards*

Typical Mistake:

When explaining the importance for standardised parts, don’t forget to include the role of organisations such as BSI in regulating compatibility.